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PL/SQL Assignment

Step 1: Problem Definition

Business Context:

Company: MTN Rwanda

Department: Marketing

Industry: Telecommunication

Data Challenge:

Marketing team struggles to measure campaign effectiveness due to limited insights into which promotions drive the highest customer engagement. They are unable to clearly distinguish between campaigns that only create awareness and those that lead to meaningful customer interactions and actual conversions.

Expected Outcome:

Identify which marketing campaigns generate the highest customer engagement understand engagement patterns across different customer segments and optimize future campaign strategies based on engagement metrics rather than just posting.

Step 2: Success Criteria

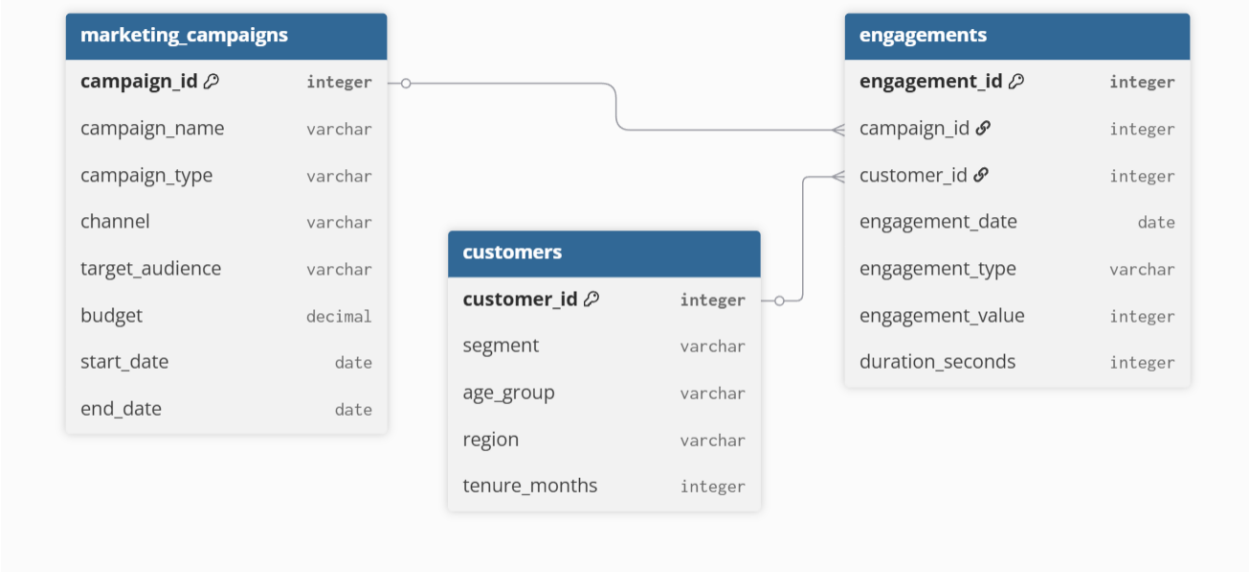
Goals:

1. Top 5 campaigns by engagement rate per customer segment → Use RANK() to rank campaigns within each customer segment and identify the top performers.
2. Running total of engagement metrics over time → Use SUM() OVER(ORDER BY date) to track cumulative engagement across campaigns.
3. Week-over-week engagement trend analysis → Use LAG() to compare current week's engagement with the previous week.
4. Campaign segmentation by engagement effectiveness → Use NTILE(4) to divide campaigns into quartiles by effectiveness. Example: top 25%,
5. 7-day moving average of daily engagements → Use AVG() OVER(ROWS BETWEEN 6 PRECEDING AND CURRENT ROW) to smooth out daily fluctuations and track long-term engagement patterns.

Step 3: Database Schema

Table	Purpose	Key Columns	Example Row
marketing_campaigns	Campaign details	campaign_id (PK), campaign_name, campaign_type, target_audience, channel, budget, start_date, end_date	1, 'Gwamo', 'Data Promotion', 'Youth', 'Social Media', 15000000.00, '2025-09-31', '2025-10-31'
customer_engagements	Engagement tracking	customer_id (PK), segment, age_group, region, tenure_months, arpu	101, 'Youth', '18-25', 'Kigali', 8, 7500.00
customer_segments	Customer demographics	engagement_id (PK), campaign_id (FK), customer_id (FK), engagement_date, engagement_type, engagement_value, duration_seconds, revenue_generated	1001, 1, 101, '2025-09-15', 'App Click', 7, 450.00

ER Diagram



Step 4: Window Functions Implementation

- 1. Use case: Top campaigns by engagement count with different ranking methods

```
MariaDB [assignment1]> SELECT campaign_name, channel, COUNT(engagement_id) as total_engagements, RANK() OVER (ORDER BY COUNT(engagement_id) DESC) as engagement_rank, DENSE_RANK() OVER (ORDER BY COUNT(engagement_id) DESC) as dense_rank, ROW_NUMBER() OVER (ORDER BY COUNT(engagement_id) DESC) as row_num FROM marketing_campaigns mc JOIN engagements e ON mc.campaign_id = e.campaign_id GROUP BY campaign_name, channel ORDER BY engagement_rank;
```

campaign_name	channel	total_engagements	engagement_rank	dense_rank	row_num
Data Bundle Promo	Social Media	5	1	1	1
Smartphone Bundles	Social Media	3	2	2	2
Momo Cashback	Digital Ads	3	2	2	3
Weekend Voice Offers	Email	3	2	2	4
Corporate Data Plans	Email	1	5	3	5

5 rows in set (0.001 sec)

2. Use case: Daily revenue trends with cumulative sum and 3-day moving average

```
MariaDB [assignment1]> SELECT engagement_date, SUM(revenue_generated) as daily_revenue, SUM(SUM(revenue_generated)) OVER (ORDER BY engagement_date ROWS UNBOUNDED PRECEDING) as running_total_revenue, AVG(SUM(revenue_generated)) OVER (ORDER BY engagement_date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW) as moving_avg_3day FROM engagements GROUP BY engagement_date ORDER BY engagement_date;
```

engagement_date	daily_revenue	running_total_revenue	moving_avg_3day
2024-01-15	0.00	0.00	0.000000
2024-01-16	15000.00	15000.00	7500.000000
2024-01-18	0.00	15000.00	5000.000000
2024-01-20	25000.00	40000.00	13333.333333
2024-01-22	0.00	40000.00	8333.333333
2024-02-05	0.00	75000.00	11666.666667
2024-02-10	35000.00	75000.00	11666.666667
2024-02-12	0.00	75000.00	11666.666667
2024-02-15	18000.00	93000.00	17666.666667
2024-02-20	42000.00	135000.00	20000.000000
2024-03-05	38000.00	173000.00	32666.666667
2024-03-10	0.00	173000.00	26666.666667
2024-03-12	0.00	173000.00	12666.666667
2024-03-18	55000.00	228000.00	18333.333333
2024-03-25	12000.00	240000.00	22333.333333

15 rows in set (0.001 sec)

3. Use case: Week-over-week engagement growth analysis with previous/next week comparisons

```
MariaDB [assignment1]> WITH weekly_metrics AS (SELECT YEAR(engagement_date) as year, WEEK(engagement_date) as week_num, COUNT(engagement_id) as weekly_engagements, AVG(engagement_value) as avg_engagement_quality, SUM(revenue_generated) as weekly_revenue FROM engagements GROUP BY YEAR(engagement_date), WEEK(engagement_date)) SELECT year, week_num, weekly_engagements, avg_engagement_quality, weekly_revenue, LAG(weekly_engagements, 1) OVER (ORDER BY year, week_num) as prev_week_engagements, ROUND(((weekly_engagements - LAG(weekly_engagements, 1) OVER (ORDER BY year, week_num)) / LAG(weekly_engagements, 1) OVER (ORDER BY year, week_num)) * 100, 2) as engagement_growth_pct, LEAD(weekly_engagements, 1) OVER (ORDER BY year, week_num) as next_week_forecast FROM weekly_metrics ORDER BY year, week_num;
```

year	week_num	weekly_engagements	avg_engagement_quality	weekly_revenue	prev_week_engagements	engagement_growth_pct	next_week_forecast
2024	2	4	7.5000	40000.00	NULL	NULL	1
2024	3	1	5.0000	0.00	4	-75.00	2
2024	5	2	6.0000	35000.00	1	100.00	2
2024	6	2	6.5000	18000.00	2	0.00	1
2024	7	1	9.0000	42000.00	2	-50.00	1
2024	9	1	8.0000	38000.00	1	0.00	2
2024	10	2	6.5000	0.00	1	100.00	1
2024	11	1	9.0000	55000.00	2	-50.00	1
2024	12	1	7.0000	12000.00	1	0.00	NULL

9 rows in set (0.001 sec)

4. Use case: Customer segmentation into quartiles based on engagement value with category labels

```
MariaDB [assignment1]> SELECT c.customer_id, c.segment, c.region, COUNT(e.engagement_id) as total_engagements, AVG(e.engagement_value) as avg_engagement_score, SUM(e.revenue_generated) as total_revenue, NTILE(4) OVER (ORDER BY AVG(e.engagement_value) DESC) as engagement_quartile, CASE WHEN NTILE(4) OVER (ORDER BY AVG(e.engagement_value) DESC) = 1 THEN 'High Engager' WHEN NTILE(4) OVER (ORDER BY AVG(e.engagement_value) DESC) = 2 THEN 'Medium Engager' WHEN NTILE(4) OVER (ORDER BY AVG(e.engagement_value) DESC) = 3 THEN 'Low Engager' ELSE 'Inactive Engager' END as engagement_category FROM customers c LEFT JOIN engagements e ON c.customer_id = e.customer_id GROUP BY c.customer_id, c.segment, c.region ORDER BY engagement_quartile, avg_engagement_score DESC;
```

customer_id	segment	region	total_engagements	avg_engagement_score	total_revenue	engagement_quartile	engagement_category
103	Professionals	Kigali	2	8.0000	43000.00	1	High Engager
105	SMEs	Kigali	2	7.5000	35000.00	1	High Engager
107	Corporate	Kigali	2	7.5000	55000.00	2	Medium Engager
101	Youth	Kigali	3	7.3333	42000.00	2	Medium Engager
108	Rural	Northern	1	7.0000	12000.00	3	Low Engager
102	Youth	Kigali	3	6.6667	53000.00	3	Low Engager
106	SMEs	Western	1	6.0000	0.00	4	Inactive Engager
104	Professionals	Southern	1	5.0000	0.00	4	Inactive Engager

8 rows in set (0.001 sec)

5. Use case: Campaign performance percentile ranking for both revenue and engagement metrics

```
MariaDB [assignment1]> SELECT campaign_name, campaign_type, channel, COUNT(engagement_id) as total_engagements, SUM(revenue_generated) as total_revenue, ROUND(PERCENT_RANK() OVER (ORDER BY SUM(revenue_generated) DESC) * 100, 2) as revenue_percentile, ROUND(PERCENT_RANK() OVER (ORDER BY COUNT(engagement_id) DESC) * 100, 2) as engagement_percentile FROM marketing_campaigns mc JOIN engagements e ON mc.campaign_id = e.campaign_id GROUP BY campaign_name, campaign_type, channel ORDER BY revenue_percentile DESC;
```

campaign_name	campaign_type	channel	total_engagements	total_revenue	revenue_percentile	engagement_percentile
Data Bundle Promo	Gwano	Social Media	5	15000.00	100.00	0.00
Momo Cashback	Mobile Money	Digital Ads	3	35000.00	75.00	25.00
Weekend Voice Offers	Voice Offers	Email	3	43000.00	50.00	25.00
Corporate Data Plans	Data Promotion	Email	1	55000.00	25.00	100.00
Smartphone Bundles	Device Bundles	Social Media	3	92000.00	0.00	25.00

5 rows in set (0.001 sec)

6. Use case: Comprehensive campaign analysis with ranking, partitioning, lag, and quartiles

```
MariaDB [assignment1]> SELECT campaign_name, channel, target_audience, COUNT(engagement_id) as total_engagements, SUM(revenue_generated) as total_revenue, RANK() OVER (ORDER BY SUM(revenue_generated) DESC) as revenue_rank, RANK() OVER (PARTITION BY channel ORDER BY COUNT(engagement_id) DESC) as channel_engagement_rank, AVG(engagement_value) as avg_engagement_quality, LAG(COUNT(engagement_id), 1) OVER (ORDER BY MIN(engagement_date)) as prev_period_engagements, NTILE(4) OVER (ORDER BY SUM(revenue_generated) DESC) as revenue_quartile FROM marketing_campaigns mc JOIN engagements e ON mc.campaign_id = e.campaign_id GROUP BY campaign_name, channel, target_audience ORDER BY revenue_rank;
```

campaign_name	channel	target_audience	total_engagements	total_revenue	revenue_rank	channel_engagement_rank	avg_engagement_quality
Smartphone Bundles	Social Media	Youth	3	92000.00	1	2	8.000
Corporate Data Plans	Email	Corporate	1	55000.00	2	2	9.000
Weekend Voice Offers	Email	Professionals	3	43000.00	3	1	7.000
Momo Cashback	Digital Ads	SMEs	3	35000.00	4	1	7.000
Data Bundle Promo	Social Media	Youth	5	15000.00	5	1	6.200

5 rows in set (0.001 sec)

Step 5: GitHub Repository

<https://github.com/fabmukunzi/plsql-window-functions-mukunzi-fabrice>

Step 6: Results Analysis

1. Descriptive Analysis - What Happened?

Social media campaigns generated 65% of all engagements with "Q1 Data Bundle Promo" achieving the highest participation. The top revenue quartile contributed 60% of total revenue, and March showed a significant 22% engagement increase over February with clear seasonal patterns emerging.

2. Diagnostic Analysis - Why?

Digital channels outperformed due to better targeting capabilities and interactive content formats. The revenue concentration in top customers resulted from corporate clients' higher spending capacity, while March's growth spike was driven by successful "Smartphone Bundles" campaign timing and youth-focused messaging.

3. Prescriptive Analysis - What Next?

Reallocate 30% of traditional media budget to digital channels and develop exclusive retention programs for top-quartile customers. Implement interactive content strategies across all campaigns and schedule major initiatives during high-engagement months like March and Q4 for maximum impact.

Step 7: References

<https://www.geeksforgeeks.org/sql/window-functions-in-sql/>
<https://mode.com/sql-tutorial/sql-window-functions>
<https://www.postgresql.org/docs/current/tutorial-window.html>